binomial therom
$(x+a)^r = \sum_{b=0}^{r} \left[\binom{r}{b} x^b a^{r-b} \right]$
pascals identity
$\binom{a}{n} = \binom{a-1}{n} + \binom{a-1}{n-1}$

Reucrsions from Binomial therom

Case 1 – Non falling sums

$$(x+a)^r = \sum_{b=0}^{r} \left[\binom{r}{b} x^b a^{r-b} \right]$$

let $a = 1$

$$(x+1)^r = \sum_{b=0}^{r} \left[\binom{r}{b} x^b \right]$$

let $x = x' - 1$

$$((x'-1)+1)^r = x^r = \sum_{b=0}^{r} \left[\binom{r}{b} (x'-1)^b \right]$$

in general

$$((x-b)+a)^r = \sum_{b=0}^{r} \left[\binom{r}{b} (x-b)^b a^{r-b} \right]$$

if b=a

$$x^r = \sum_{b=0}^{r} \left[\binom{r}{b} (x-a)^b a^{r-b} \right]$$

we can subtitute the defintion of x^r into $(x-a)^b$ or a^{r-b}
$(x-a)^b$ is a good substitution as it the nested power term will shrink by a
after $\frac{x}{a}$ expansions the inner power term will be 0

$$x^r = \sum_{b=0}^{r}\left[\binom{r}{b}(x-a)^b a^{r-b}\right]$$

$$(x-a)^b = \sum_{c=0}^{b}\left[\binom{b}{c}((x-a)-a)^c a^{b-c}\right]$$

$$x^r = \sum_{b=0}^{r}\left[\binom{r}{b}\sum_{c=0}^{b}\left[\binom{b}{c}((x-a)-a)^c a^{b-c}\right]a^{r-b}\right]$$

every layer strips one a from inner power's base

$$x^r = \sum_{b=0}^{r}\left[\binom{r}{b}\sum_{c=0}^{b}\left[\binom{b}{c}((x-a)-a)^c a^{b-c}\right]a^{r-b}\right]$$

$$x^r = \overbrace{\sum_{b=0}^{r}\left[\binom{r}{b}\sum_{c=0}^{b}\left[\binom{b}{c}\sum_{d=0}^{c}\left[\binom{c}{d}\cdots\left[\sum_{n_{\frac{x}{a}}=0}^{n_{(\frac{x}{a})-1}}\left[\binom{n_{(\frac{x}{a})-1}}{n_{\frac{x}{a}}}0^{n_{\frac{x}{a}}}a^{n_{(\frac{x}{a})-1}-n_{\frac{x}{a}}}\right]\right]a^{c-d}\right]a^{b-c}\right]a^{r-b}\right]}^{\frac{x}{a}\text{ sums}}$$

you can stop at any layer you want, you dont need to fully expand

$$x^r = \overbrace{\sum_{b=0}^{r}\left[\binom{r}{b}\sum_{c=0}^{b}\left[\binom{b}{c}\sum_{d=0}^{c}\left[\binom{c}{d}\cdots\left[\sum_{n_l=0}^{n_{l-1}}\left[\binom{n_{l-1}}{n_l}(x-la)^{n_l}a^{n_{l-1}-n_l}\right]\right]a^{c-d}\right]a^{b-c}\right]a^{r-b}\right]}^{l\text{ sums}}$$

Case 2 – falling sums

$$(x+a)^r = \sum_{b=0}^{r}\left[\binom{r}{b}x^b a^{r-b}\right]$$

$$(x+a)^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}x^b a^{r-b}\right] + \binom{r}{r}x^r a^0$$

$$((x-a)+a)^r = x^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}(x-a)^b a^{r-b}\right] + \binom{r}{r}(x-a)^r a^0$$

the same reassoning applys here as in the last case:
"we can subtitute the defintion of x^r into $(x-a)^b$ or a^{r-b}
$(x-a)^b$ is a good substitution as it the nested power term will shrink by a
after $\frac{x}{a}$ expansions the inner power term will be 0 "

$$x^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}(x-a)^b a^{r-b}\right] + \binom{r}{r}(x-a)^r a^0$$

$$(x-a)^b = \sum_{c=0}^{b-1}\left[\binom{r}{b}((x-a)-a)^c a^{b-c}\right] + \binom{b}{b}((x-a)-a)^b a^0$$

$$x^r = \sum_{b=0}^{r-1}\left[\binom{r}{b}\left[\sum_{c=0}^{b-1}\left[\binom{r}{b}((x-a)-a)^c a^{b-c}\right] + \binom{b}{b}((x-a)-a)^b a^0\right]a^{r-b}\right] + \binom{r}{r}(x-a)^r a^0$$

$$x^r = \overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}\left[\sum_{c=0}^{b-1}\left[\binom{r}{b}\right.\right.\right.}^{l \text{ sums}}$$

$$\left.\left.\left.\cdots\left[\sum_{n_l=0}^{n_{l-1}-1}\left[\binom{n_{l-1}}{n_l}(x-la)^{n_l}a^{n_{l-1}-n_l}\right] + \binom{n_{l-1}}{n_{l-1}}(x-la)^{n_{l-1}}a^0\right]\cdots\right.\right.\right.$$

$$\left.\left.\left.a^{b-c}\right] + \binom{b}{b}(x-2a)^b a^0\right]a^{r-b}\right] + \binom{r}{r}(x-a)^r a^0$$

$$a^0 = 1 \text{ for all } a \in \mathbb{R}$$

$\binom{r}{r} = 1$ for all $r \in \mathbb{Z}^+$

if we restrict our formula with these we can simplify it

according to our restriciton $a^0 = 1$ and $\binom{r}{r} = 1$

$$x^r = \overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}\left[\sum_{c=0}^{b-1}\left[\binom{r}{b}\cdots\left[\sum_{n_l=0}^{n_{l-1}-1}\left[\binom{n_{l-1}}{n_l}(x-la)^{n_l}a^{n_{l-1}-n_l}\right] + (x-la)^{n_{l-1}}\right]\cdots a^{b-c}\right] + (x-2a)^b\right]a^{r-b}\right] + (x-a)^r}^{l \text{ sums}}$$

the sum can be factored into a sum of falling sums

$$x^r = \overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}\left[\sum_{c=0}^{b-1}\left[\binom{r}{b}\cdots\left[\sum_{n_l=0}^{n_{l-1}-1}\left[\binom{n_{l-1}}{n_l}(x-la)^{n_l}a^{n_{l-1}-n_l}\right]\right]\cdots a^{b-c}\right]\right]a^{r-b}\right]}^{l \text{ sums}}$$

$$+\overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}\left[\sum_{c=0}^{b-1}\left[\binom{r}{b}\cdots\left[\sum_{n_{l-1}=0}^{n_{l-2}-1}\left[(x-la)^{n_{l-1}}\right]\right]\cdots a^{b-c}\right]\right]a^{r-b}\right]}^{l-1 \text{ sums}}$$

$$+\ldots+\overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}\left[(x-2a)^b\right]a^{r-b}\right]}^{1 \text{ sums}}$$

$$+\overbrace{(x-a)^r}^{0 \text{ sums}}$$

$$x^r = \overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}\left[\sum_{c=0}^{b-1}\left[\binom{r}{b}\ldots\left[\sum_{n_l=0}^{n_{l-1}-1}\left[\binom{n_{l-1}}{n_l}(x-la)^{n_l}a^{n_{l-1}-n_l}\right]\right]\ldots a^{b-c}\right]\right]a^{r-b}\right]}^{l \text{ sums}}$$

$$+\overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}\left[\sum_{c=0}^{b-1}\left[\binom{r}{b}\ldots\left[\sum_{n_{l-1}=0}^{n_{l-2}-1}\left[(x-la)^{n_{l-1}}\right]\right]\ldots a^{b-c}\right]\right]a^{r-b}\right]}^{l-1 \text{ sums}}$$

$$+\ldots+\overbrace{\sum_{b=0}^{r-1}\left[\binom{r}{b}\left[(x-2a)^b\right]a^{r-b}\right]}^{1 \text{ sums}}$$

$$+\overbrace{(x-a)^r}^{0 \text{ sums}}$$

Source: ./formulas.typ (<eq:case2-l-sums>)